C. Non-coprime Split

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: math, number theory | https://codeforces.com/problemset/problem/1872/C

You are given two integers $l \le r$. You need to find **positive** integers a and b such that the following conditions are simultaneously satisfied:

- $l \le a + b \le r$
- $\gcd(a, b) \neq 1$

or report that they do not exist.

$\gcd(a, b)$ denotes the greatest common divisor of numbers a and b. For example, $\gcd(6, 9) = 3$, $\gcd(8, 9) = 1$, $\gcd(4, 2) = 2$.

Input

The first line of the input contains an integer t ($1 \le t \le 500$) — the number of test cases.

Then the descriptions of the test cases follow.

The only line of the description of each test case contains 2 integers l, r ($1 \le l \le r \le 10^7$).

Output

For each test case, output the integers a, b that satisfy all the conditions on a separate line. If there is no answer, instead output a single number -1.

If there are multiple answers, you can output any of them.

Example

Input
11 11 15 1 3 18 19 41 43 777 777 8000000 10000000 2000 2023 1791791 1791791 1 4 2 3 9840769 9840769

Output

```
6 9
-1
14 4
36 6
111 666
4000000 5000000
2009 7
-1
2 2
-1
6274 9834495
```

Note

In the first test case, $11 \le 6 + 9 \le 15$, gcd(6, 9) = 3, and all conditions are satisfied. Note that this is not the only possible answer, for example, {4, 10}, {5, 10}, {6, 6} are also valid answers for this test case.

In the second test case, the only pairs {a, b} that satisfy the condition $1 \le a + b \le 3$ are {1, 1}, {1, 2}, {2, 1}, but in each of these pairs gcd(a, b) equals 1, so there is no answer.

In the third sample test, gcd(14, 4) = 2.